

TrES-3b

R-0041

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260417 · created 2026-07-09T00:55:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461147.8869 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.058
Transit depth (Rp/Rs)^2	3.81 +/- 2.27 [%]
Orbital Inclination (inc)	84.08 +/- 2.91
Airmass coefficient 1 (a1)	1.11 +/- 0.59
Airmass coefficient 2 (a2)	-0.16 +/- 0.55
Scatter in the residuals of the lightcurve fit is	50.27 %
Best Comparison Star	None
Optimal Aperture	1.27
Optimal Annulus	4.4
Transit Duration (day)	0.066 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.058 vs 0.16309 (deviation 0.52σ)
Duration fit vs published	0.066 d vs 0.0567 d (deviation 0.42σ)
Mid-time O–C	3.3 min
Depth SNR	3.2
Residual scatter	50.27 %

Light curve

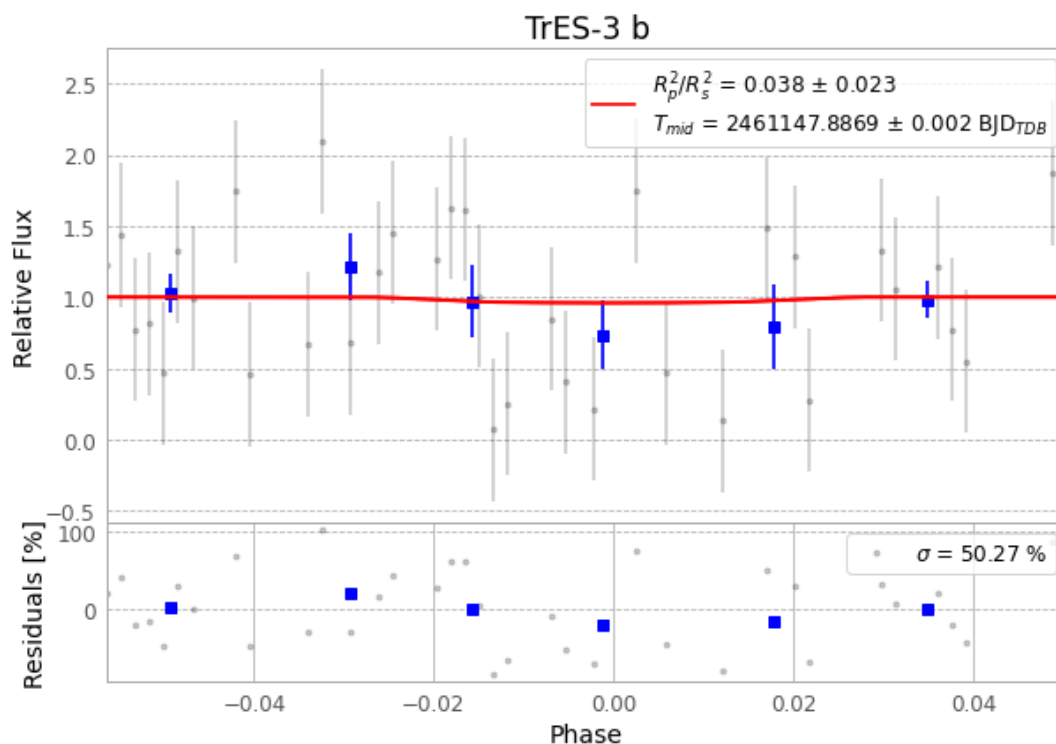


Figure 1 — FinalLightCurve_TrES-3 b_2026-04-17.png (SHA-256 875834dfd19cdce6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 26549d7516058dc7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

70 FITS frames (46 MB), TRES-3260417072718.FITS → TRES-3260417105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260417.log (e82e5c415f83...), FinalLightCurve_TrES-3 b_2026-04-17.png (875834dfd19c...), FinalLightCurve_TrES-3 b_2026-04-17.csv (9febf75bc1dc...)

Compute resources

Wall time 127.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.